

Natural Language Processing

Part 2 – Project Instruction

Organizational Details

- Requirements for ECTS credits + final assessment:
- Group work (approx. 3–4 people per group):
 - Each group comes up with a research question and a topic they would like to work on.
 - We will always have enough time to discuss and answer questions.
 - If you have questions, please use the Moodle forum.
- Final assessment (50% of the total semester grade → 2.5 ECTS):
 - Group presentation in the final session! We have 3 hours for this (50%).
 - Code submission via Moodle per group (50%) → Python!

Proposal Submission: 31.05.2026

Project Presentation: 01.07.2026 and 02.07.2026

Originality Requirement

- Your project must go beyond simply reproducing an existing online project (e.g., tutorials, GitHub repositories, blog posts). You may use such resources only as a starting point, but you must:
 - Clearly cite all sources
 - Modify or extend the approach in a meaningful way
 - Demonstrate independent thinking and decision-making
- Projects that closely mirror existing work without meaningful changes will receive low marks.

Problem Definition

- Clearly define:
 - The task (e.g., classification, summarization, NER, etc.)
 - The dataset (public or self-collected)
 - The motivation and expected challenges

Baseline Model (Required)

- You must implement at least one baseline approach, such as:
 - Simple machine learning model (e.g., logistic regression, Naive Bayes)
 - Basic neural model
- This serves as a reference point for improvement.

Experimental Design (Core Requirement)

- You are required to perform multiple experiments. Simply reporting a single model result - even if performance is low - is not sufficient.
- Your experiments must include:
 - At least 3 meaningful variations, such as:
 - Different model architectures
 - Feature engineering choices
 - Preprocessing strategies
 - Hyperparameter tuning
 - Data augmentation
 - A clear hypothesis for each experiment
 - A comparison of results across experiments

Performance and Iteration

- If your initial results are poor, you are expected to:
 - Analyze why performance is low
 - Propose and test improvements
 - Show an iterative process
- Good projects demonstrate progress and reasoning, not just final performance.

Interpretation and Analysis (Critical)

- You must include interpretability and analysis, such as:
 - Error analysis (What kinds of mistakes does the model make?)
 - Model behavior insights (biases, limitations)
 - Visualization (attention maps, confusion matrices, etc.)
 - Explanation of why certain approaches worked or failed
- Projects with strong interpretation will be graded higher than those with only raw results.

Report Structure

- Your final presentation must include:
 1. Introduction & Problem Definition
 2. Related Work / Inspiration
 3. Methodology
 4. Experiments
 5. Results
 6. Analysis & Interpretation
 7. Conclusion & Future Work

Deliverables

- Code (well-documented and reproducible)
- Final presentation (PDF)
- Optional: short presentation or demo

Grading Criteria

- Originality and creativity
- Experimental depth and rigor
- Quality of analysis and interpretation
- Clarity of explanation
- Technical correctness

Research Topics

1. Extend the development of an end-to-end ESG (Environmental, Social, and Governance) knowledge graph from alternative data (using LLMs for relationship extraction and GNNs for latent relationship detection).
2. Build a domain-specific knowledge graph to improve legal document analysis in natural language processing applications.
3. Design an interactive knowledge graph for enhanced patient information management in healthcare chatbots.
4. Implement a knowledge graph to optimize content recommendation systems in media streaming platforms.
5. Develop a knowledge graph to analyze and visualize ESG data for sustainable investment strategies.
6. Develop debiasing strategies for large language models to enhance gender neutrality in job description generation.
7. Apply debiasing methods to improve ethnic and racial fairness in sentiment analysis using large language models.
8. Enhance entity recognition in textual data using graph neural networks for improved information retrieval.
9. Develop advanced topic modeling techniques to discover emerging trends in social media content.
10. Apply topic modeling to enhance content curation and personalization in news aggregation platforms.